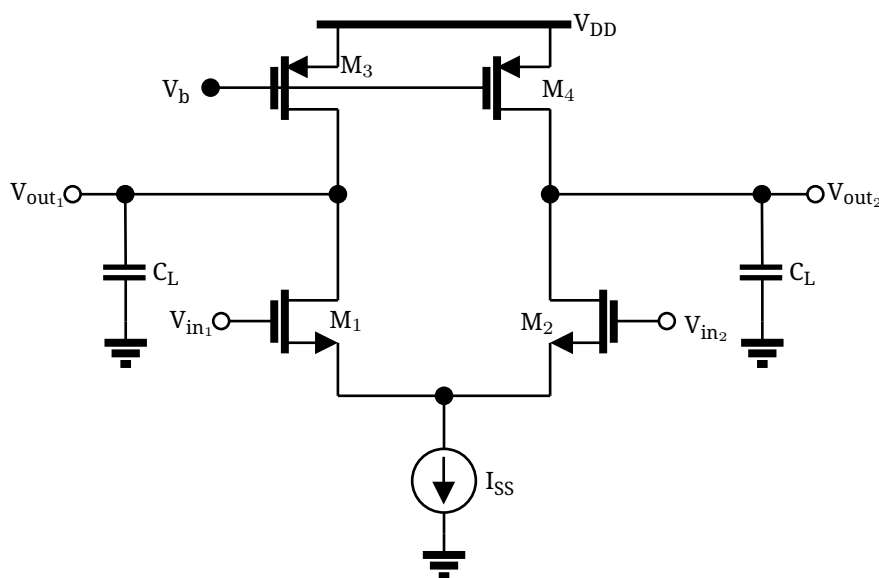


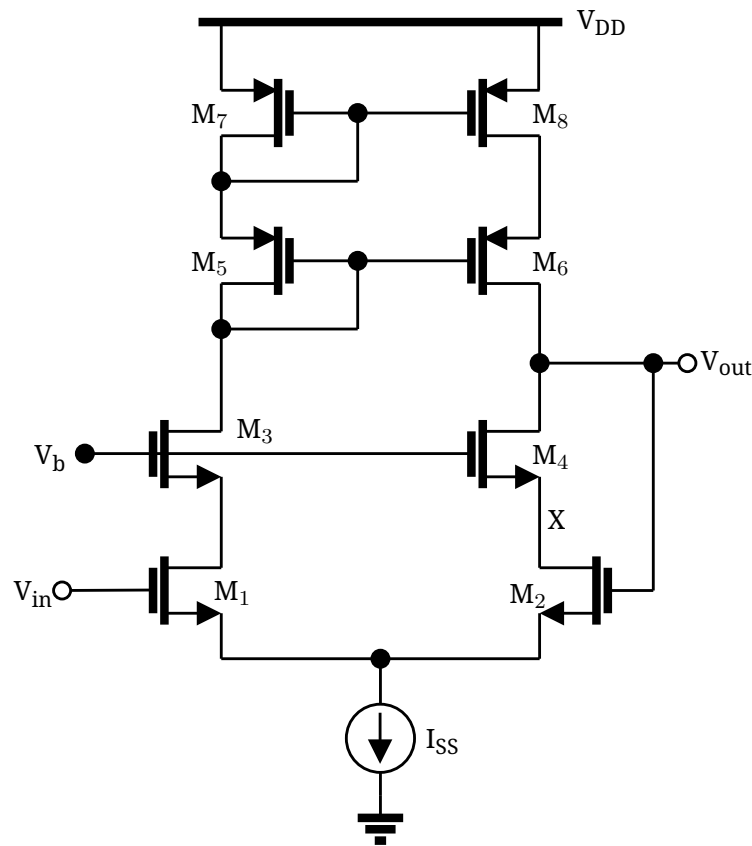
Use following parameters for this tutorial:

$\mu_n C_{ox} = 134.28 \mu A/V^2$ ,  $\mu_p C_{ox} = 38.36 \mu A/V^2$ ,  $\lambda_n = 0.1 V^{-1}$  for  $L=0.5 \mu m$ ,  $\lambda_p = 0.2 V^{-1}$  for  $L=0.5 \mu m$ ,  $V_{T0,n}=0.7 V$ ,  $V_{T0,p}=-0.8 V$ ,  $V_{DD}=3 V$

1. (a) Derive expressions for the transconductance and output resistance of a MOSFET in the triode region.
- (b) Consider the amplifier of figure below, with  $(W/L)_{1-4} = 50/0.5$ ,  $I_{SS} = 1 mA$ , and input CM level of 1.3 V. Calculate the small-signal gain and the maximum output swing if all transistors remain in saturation.
- (c) For the circuit of part (b), suppose we allow each PMOS device to enter the triode region by 50 mV so as to increase the allowable differential swing by 100 mV. What is the small-signal gain at the peaks of the output swing?



2. In the circuit of figure below, assume that  $(W/L)_{1-4} = 100/0.5$ ,  $I_{SS} = 1 mA$ ,  $V_b = 1.4 V$ , and  $\gamma = 0$ .
  - (a) If  $M_3 - M_8$  are identical and have a length of  $0.5 \mu m$ , calculate their minimum width such that  $M_3$  operates in saturation.
  - (b) Calculate the maximum output voltage swing.
  - (c) What is the open-loop voltage gain?



3. Design the folded-cascode op amp of figure below for the following requirements: maximum differential swing = 2.4 V, total power dissipation = 6 mW. If all of the transistors have a channel length of 0.5  $\mu\text{m}$ , what is the overall voltage gain? Can the input common-mode level be as low as zero?

